

Catching Our Own Errors: Clean Initialization and Blind Track Skill in a Tropical Cyclone Model

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Capsule. A tropical-cyclone model initialized from observations with no track-fitted parameter reproduces six landfalls and exposes an intrinsic, over-rotating β -gyre—a model error characterized in the open rather than hidden behind compensating fudge factors.

Abstract. A landfall track error convolves a storm’s self-propagation, its environmental steering, and whatever track features a model cannot represent; landfall data alone cannot separate them, and the standard practice of developing a model against the hindcasts it is judged on selects for errors that cancel. We describe a discipline, *clean initialization*, under which a tropical-cyclone model is denied that opportunity: every storm is initialized directly from the HURDAT2 best track, all storms run under one storm-agnostic configuration, and the single structural free parameter is calibrated against canonical β -drift physics in a quiescent testbed—never against any landfall. The discipline produced two results and a falsification. First, a characterized model property: the model’s β -gyres over-rotate, yielding self-propagation of canonical magnitude aimed nearly due poleward, a bias that survives structure, diffusion, and resolution levers and an f -plane null. Second, blind track skill: six historical Atlantic landfalls—three never run during development—are reproduced with cross-track errors from a few tens of kilometers to roughly 125 km. The obvious bridge between the two facts—that the bias causes the landfall errors—was drafted, tested by projecting the bias through each storm’s landfall geometry, and falsified: steering controls landfall cross-track, and the bias is a subdominant along-track term. We recount the compensating-errors cascade the discipline exposed—seven errors, each hidden behind the one above it—and offer the method as a template for trustworthy model evaluation.

Significance Statement. Weather models are usually adjusted until they reproduce past events—a practice that can hide errors that happen to cancel. We built a hurricane track model that is never tuned to observed tracks: it is initialized from the official hurricane database, and its one structural setting is calibrated against textbook vortex physics. Run on six historical hurricanes—three of them never used during development—it reproduced landfall positions without any storm-specific adjustment. Just as important, the discipline exposed a specific, measurable error in how the model’s storms steer themselves, and a chain of hidden, compensating errors in our own physics, data, and measurement tools that conventional tuning would have buried. We offer the approach as a template for trustworthy model evaluation.

1 Introduction

This paper reports two facts about one tropical-cyclone model—a characterized error and a blind success—and the test that stopped us from claiming the obvious bridge between them.

The facts concern track. A tropical cyclone’s motion is, to leading order, a superposition of two influences: *environmental steering*, the storm carried along by the surrounding deep-layer flow, and *self-propagation*, the storm pushing itself. The classical self-propagation mechanism is β -drift: a vortex on the rotating Earth organizes a pair of counter-rotating gyres out of the planetary-vorticity gradient, and the flow between those gyres propels the storm—canonically toward the northwest, at $1\text{--}3\text{ m s}^{-1}$, for a Northern Hemisphere cyclone. Neither influence is directly observable in a track; what is observable, at the moment of landfall, is their sum. Throughout this paper we decompose track error into components along and across the observed track: the along-track component is a timing error—the storm arrives early or late—while the cross-track component displaces landfall itself, and with it every downstream judgment that depends on where a storm comes ashore.

That decomposition names an attribution problem it cannot by itself solve. A landfall error is a convolution of self-propagation error, steering error, and whatever track features the model cannot represent at all; landfall data alone cannot separate them, and a compensating pair of errors in any two produces a small landfall miss for free. The standard development practice—adjust the model until the hindcast matches—is, seen from this angle, a machine for manufacturing exactly such cancellations. Each tuning step is accepted because it reduces the error against the cases in hand; nothing in that procedure distinguishes a model that is right from a model whose errors have been arranged to cancel, and the arrangement fails silently on the first storm that breaks its geometry. Section 3 argues that this hazard is not hypothetical: we document a cascade in our own development in which seven successive errors—in the physics, the configuration, the input data, and finally our measurement instruments—each hid behind a result that looked good.

Our response is a discipline we call *clean initialization*. Every storm is initialized directly from the HURDAT2 best track, with the initialization values verified against the primary source file. Every storm runs under one storm-agnostic configuration. The configuration’s single structural free parameter—the radius at which the initial vortex’s outer wind profile begins to taper—is calibrated in a quiescent-environment testbed against the canonical β -drift range from the published literature, never against any landfall. And residuals are reported as findings rather than absorbed into the configuration. The cost of this discipline is that the model misses visibly, including in ways a tuned model would not. The payoff is that its misses mean something—and so do its hits.

The two facts are as follows. First, clean initialization exposed a structural error worth characterizing in its own right: the model’s β -gyres *over-rotate*. In the isolated testbed the simulated β -drift has canonical magnitude but is aimed nearly due poleward—its westward component is a fraction of the canonical value—and the deficit survives every lever that could plausibly control it: outer-vortex structure, diffusion, a doubling of horizontal resolution, and an f -plane null test that rules out numerical and tracker artifacts (Section 4.1). It is a bounded, reproducible, mechanistically located bias in self-propagation aim, and we report it as a model property rather than removing it. Second, the same configuration exhibits blind track skill: six historical Atlantic landfalls—Hugo (1989), Katrina (2005), and Ivan (2004), on which the residual was first characterized, and Fran (1996), Michael (2018), and Laura (2020), which the configuration had never run—are reproduced with cross-track errors ranging from a few tens of kilometers to roughly 125 km, with no track-fitted parameter anywhere in the system (Section 4.2).

The obvious paper connects these facts: the discovery storms all landed east of their observed tracks by a strikingly consistent $\sim 120\text{ km}$, and an over-rotating β -gyre pushes poleward-moving storms east. We drafted that paper. It was wrong, and the manner of its failure is the third thing this paper reports. The three test storms broke the cluster—two of them missed *west*—and a projection test showed that no single self-propagation bias vector, pushed through each storm’s

landfall geometry, can reproduce the six observed errors: two storms approaching on the same heading missed on opposite sides. Steering controls landfall cross-track in these cases; the β -gyre bias is real, detectable—it predicts one test storm’s along-track error almost exactly—but subdominant (Section 4.2). We therefore present the two results as two results, with the honest relation between them, and we present the methodology that forced that honesty—the compensating-errors cascade, the registered predictions, the claims retired by our own probes before review—as a contribution in its own right (Section 3), of particular relevance as model development workflows become increasingly automated.

The remainder of the paper is organized as follows. Section 2 describes the model—a nonhydrostatic anelastic core run, deliberately, in a reduced barotropic configuration—and the experimental design: initialization, the single calibrated parameter, steering, and track verification. Section 3 presents the clean-initialization methodology and the cascade of compensating errors it exposed. Section 4 presents the two results: the over-rotating β -gyre as a characterized model property (4.1) and the six-storm landfall record with its along/cross attribution (4.2). Section 5 discusses what the over-rotation is and is not, the limits of landfall attribution, and what blind skill under a no-tuning discipline does and does not demonstrate. Section 6 concludes.

2 Dynamical Core and Experimental Design

2.1 Dynamical Core

Oracle V8 is a three-dimensional, nonhydrostatic dynamical core solving the Lipps and Hemler (1982) anelastic equations—the standard soundproof system for deep convection, in which acoustic modes are filtered by the mass-continuity constraint $\nabla \cdot (\bar{\rho}\mathbf{u}) = 0$ about a hydrostatic, stably stratified base state. The prognostic variables are the three velocity components and the potential-temperature perturbation θ' ; pressure enters as the potential of a projection that enforces the constraint. The domain is doubly periodic in the horizontal with a rigid surface and lid, on a β -plane whose reference Coriolis parameter is set from each storm’s initialization latitude. The operating grid spacing is $\Delta x = 15.6$ km with 32 levels over a 20-km depth, a 30-s time step, and per-storm domains of 4000–8000 km chosen by a fixed geometric rule (Section 2.2) so that the storm’s circulation never approaches the domain edges.

One configuration choice must be stated prominently, because the results of Section 4 are read against it. For the track experiments in this paper, the thermodynamic pathway is switched off: θ' is zeroed at initialization and its buoyancy forcing is not applied, so temperature perturbations are dynamically passive and the track dynamics reduce to vortex self-advection, the β -effect, imposed environmental steering, and boundary-layer friction. In practice, the configuration is barotropic—deliberately. Track at these scales is a steering-plus- β -drift problem; running the core in its reduced configuration makes the β -gyre characterization of Section 4.1 clean, and directly comparable to the idealized barotropic β -drift literature (Chan and Williams 1987; Fiorino and Elsberry 1989) and to operational barotropic track models of the BAM/VICBAR class. What distinguishes Oracle from those models is that the reduction is a *configuration*, not an *architecture*: the same core runs fully thermodynamically active, and the anelastic system’s foundational small-perturbation assumption has been validated in that mode. The dry, buoyancy-off track configuration is the honest minimum that the two results of this paper require.

The remaining ingredients are the model’s dissipative and bounding operators—scale-selective ∇^4 hyperdiffusion, a vorticity-preserving divergence damper, bulk-aerodynamic surface drag, Newtonian

cooling, and an intensity ceiling. All coefficients are fixed across every storm in this paper; none is adjusted per storm or per landfall (Section 2.2).

Inside the Dynamical Core

Equations and base state. The LH82 anelastic system prognoses (u, v, w, θ') about a dry base state with constant buoyancy frequency $N = 0.01 \text{ s}^{-1}$ ($\bar{\theta} = 300 \text{ K} \cdot \exp(N^2 z/g)$; $\bar{\rho}$ from discrete hydrostatic integration of the Exner function). Buoyancy, when active, enters the vertical momentum equation as $b = g \cdot \theta' / \bar{\theta}$. The equation set is implemented as a swappable abstraction—the pseudo-incompressible system of Durran (1989) is the designed alternative—so that the model can interrogate its own foundational approximation. That approximation, $|\theta'| \ll \bar{\theta}$, was tested directly in a buoyancy-enabled eyewall configuration with prescribed annular heating: the terms LH82 neglects grow linearly at $\approx 0.85 \times \theta' / \bar{\theta}$ and remain at the few-percent level for realistic eyewall amplitudes ($\theta' / \bar{\theta} \approx 4\%$), with stable integration to $\theta' / \bar{\theta} \approx 9\%$ and no sign of qualitative breakdown.

Grid and time stepping. Vertical staggering is the Lorenz grid (u, v, θ' , and the projection potential on level centers; w on level interfaces), with $w = 0$ at the rigid surface and lid. Horizontal boundary conditions are periodic. Time integration is the three-stage Runge–Kutta of Wicker and Skamarock (2002), with slow tendencies (advection, Coriolis, drag, diffusion, damping, cooling) applied as Strang-split half-steps around the buoyancy-driven fast stages; in the barotropic track configuration the fast stages are dormant and each step reduces to the split slow dynamics. Advection is second-order centered in the advective form. The anelastic constraint is re-enforced after every velocity-modifying substep by a projection: the variable-coefficient Poisson problem $\nabla \cdot (\bar{\rho} \nabla \phi) = \nabla \cdot (\bar{\rho} \mathbf{u}^*)$ is solved by 2-D FFT in the horizontal and a tridiagonal (Thomas) solve in each wavenumber column, with Neumann conditions at surface and lid and the gauge pinned at the mean mode. The solvability (compatibility) residual of the singular mean mode is logged at every solve and sits at machine precision across full multi-day integrations—a running certificate that the projection is well posed, not merely stable.

Stabilization and bounding operators (storm-agnostic values in parentheses).

- *Hyperdiffusion*: biharmonic ∇^4 on momentum ($\nu_4 = 3 \times 10^{11} \text{ m}^4 \text{ s}^{-1}$), scale-selective by design—its damping time is ~ 20 minutes at the $2\Delta x$ scale but $\sim$ days at the vortex scale, so grid noise is removed without eroding the storm.
- *Divergence damping*: the divergent part of the horizontal flow is isolated by a Helmholtz decomposition (an FFT Poisson solve) and relaxed ($\epsilon = 0.5$ per half-step); because the correction is a pure gradient it adds exactly zero vorticity, leaving the balanced circulation untouched.
- *Surface drag*: bulk-aerodynamic ($C_d = 1.5 \times 10^{-3}$), decaying linearly to zero at $H_{bl} = 1 \text{ km}$.
- *Newtonian cooling*: θ' relaxation ($\tau = 30 \text{ min}$) bounding adiabatic temperature anomalies, the standard dry-model surrogate for radiative equilibration (Emanuel 1986).
- *Intensity ceiling*: perturbation winds are relaxed toward a 70 m s^{-1} cap ($\tau = 300 \text{ s}$), a guard against numerical intensity runaway (Section 3.2, chapter 1).

The β -plane's $f(y)$ is tapered smoothly back to f_0 over the outer 20% of the domain at the north and south boundaries so that f is continuous across the periodic seam; the interior 60% is an exact β -plane, and the domain rule of Section 2.2 keeps every storm inside it. Coriolis, drag, and the intensity cap all act on the *departure* of the wind from the imposed environmental steering flow, which is treated as a maintained geostrophic background.

2.2 Vortex Initialization and the Single Structural Parameter

Every storm is initialized directly from the HURDAT2 best-track file (Landsea and Franklin 2013) at the chosen initialization time. Position, maximum wind, central pressure, and the Coriolis parameter (set from the initialization latitude) are read from the $t = 0$ fix by the loader—no initialization value is hand-entered—and the verification reference is simply every subsequent best-track fix (Section 2.4). The file itself, and our reading of it, were verified against the current and prior HURDAT2 editions; Section 3.2 recounts what that verification caught.

The initial vortex is a Holland (1980) gradient-wind-balanced cyclone, $V(r) = V_{\max}(R_{\max}/r)^B \exp[1 - (R_{\max}/r)^B]$, with the observed maximum wind, a shape parameter $B = 1.5$ frozen across all storms, and $R_{\max}$ fixed at 75 km for every storm—a documented resolution floor ($\approx 5\Delta x$ at the operating grid spacing), not observed inner-core structure. The profile decays exponentially with height from its surface reference; pressure follows from gradient-wind balance, integrated inward from the environmental radius. (The associated warm-core temperature perturbation is then zeroed, per the barotropic configuration of Section 2.1). Five projection-only iterations remove residual initialization divergence before the integration clock starts.

Left untreated, the Holland profile decays so slowly with radius ($r^{-B/2}$) that the circulation fills any domain, and the storm’s β -drift inflates accordingly (Section 3.2, chapter 3). The outer wind is therefore tapered to zero by a cosine ramp beginning at the *taper-onset radius* and completing at the environmental radius $R_{\text{env}} = 500$ km. The taper-onset radius is the configuration’s **single structural free parameter**—and the sweeps show the two nominal knobs are degenerate (only the product of R_{env} and the onset fraction matters), so it is genuinely one parameter, not two. It was calibrated in the quiescent-environment testbed of Section 4.1: sweeping the onset radius moves the simulated β -drift magnitude and aim together, and an onset radius of 200 km ($\approx 2.7R_{\max}$) brings the drift magnitude into the canonical published range while minimizing the poleward aim bias. That value was locked before any production storm run and shared by every storm in this paper. We emphasize the resulting property on which the “blind” in blind track skill rests: **no parameter anywhere in the system has ever been adjusted against a landfall**. The calibration target is idealized-vortex physics from the literature, and no landfall-fitting pathway exists in the code.

Domain size and run length are likewise derived, not chosen: a fixed geometric rule takes each storm’s initialization and threshold latitudes and returns the smallest standard domain (4000–8000 km; 256–512 grid points at fixed Δx) that keeps the storm’s full circulation, with margin, inside the exact- β interior of Section 2.1’s tapered β -plane; integrations run 10 h past the observed landfall time.

2.3 Environmental Steering

The environment is imposed as a horizontally uniform background flow relaxed toward the ERA5 (Hersbach et al. 2020) deep-layer mean (DLM)—the mass-weighted 850/700/500/300-hPa average, the standard tropical-cyclone steering layer—computed on the raw ERA5 grid and averaged over a 3–7° annulus centered on the *model* storm, so that the inner core’s own circulation does not steer it. The annulus is sampled at the model’s position and interpolated in time: the storm feels the environment where it actually is, not where the observed storm was. Every 30 minutes of simulation the background flow relaxes toward the locally sampled DLM with a 3-h timescale; each increment is applied simultaneously to the flow carried in the model state and to the perturbation references of the Coriolis, drag, and intensity-cap operators (the perturbation-relative convention of Section 2.1), so the forcing operators and the flow they reference advance in lockstep. Because the increment

is horizontally uniform it carries no divergence and no vorticity: the projection, and the vortex itself, are untouched by construction. Time-varying steering is architecture here, not refinement: in a configuration accident that became an informative A/B test, steering frozen at its initial value failed in *opposite* directions on different storms (+182 km east on Hugo, −192 km west on Ivan), which no steering-independent mechanism can produce. One honest limitation is deferred to Section 5: the annulus average is not a storm-removed environmental field, and at these radii it retains some storm-induced flow.

2.4 Track Verification

All track scoring passes through one shared verification module, against the HURDAT2 fixes, in two layers. The first is the headline same-latitude threshold comparison: the model’s and the observed track’s first northward crossings of a threshold latitude (the observed landfall latitude) are interpolated, and the crossing-time difference and the longitude offset at crossing separate timing from cross-track displacement. The second is the full along/cross decomposition evaluated at every observed fix: the along-track component is the model’s displacement ahead of (or behind) the observed storm along its instantaneous direction of motion, and the cross-track component is the displacement to the right of that motion—approximately eastward for the poleward-moving storms considered here. Two retired metrics motivate this convention. The legacy landfall-point comparison—model threshold crossing versus the observed landfall point—folds the observed storm’s remaining along-track travel into apparent cross-track error and overstated one re-scored run by half. And any single cross-track scalar can manufacture error for a storm that is not moving due north at threshold: Michael’s poleward along-track overshoot against an observed track recurving northeast projects onto the same-latitude axis as a spurious *westward* miss, which the full decomposition correctly renders as timing (Sections 3.3, 4.2). Section 4 therefore reports both layers for all six storms.

3 Clean Initialization and the Compensating-Errors Cascade

3.1 The Principle

A landfall hindcast can be right for the wrong reasons. A storm’s track error at landfall is a convolution of at least three sources—the storm’s self-propagation, the imposed environmental steering, and track features the model cannot represent—and a pair of compensating errors in any two of them will produce a small landfall miss for free. Worse, when a model is developed against the same landfalls it is judged on, such cancellations are not merely possible but *selected for*: every tuning step migrates the configuration toward settings where errors offset, and the model’s failure is silently deferred to the first storm where the cancellation does not hold. Nothing in the final error statistics distinguishes a model that is right from one whose errors are balanced.

We therefore adopted a discipline we call *clean initialization*:

- Every storm is initialized directly from the HURDAT2 best track, with the initialization values themselves verified against the primary source file.
- All storms run under one storm-agnostic configuration, with no per-storm adjustments of any kind.
- The configuration’s single structural free parameter—the radius at which the outer wind profile begins to taper—is calibrated in a quiescent-environment testbed against the canonical

β -drift range from the published literature (Section 2.2), never against any landfall.

- Residuals are reported as findings, not absorbed into the configuration.

Under this discipline the model has nowhere to hide an error. The corollary, which this section illustrates, is that development proceeds by *exposing* errors—including several that flattering early results had concealed, and several introduced by our own tooling and data handling.

3.2 The Cascade

It is one thing to state the hazard of compensating errors and another to watch it operate. The model’s development ran as a cascade: seven times, repairing one error exposed the next error it had been hiding. We recount the sequence because the sequence is the point—no single audit, however careful, would have found the seventh error while the first six stood in front of it.

1. **A numerical intensity runaway.** Early integrations developed grid-scale perturbation winds exceeding 150 m s^{-1} —numerical growth, not intensification. A bounded relaxation toward a 70 m s^{-1} ceiling contained it, and simulated intensities became honest. Only then was the track readable at all.
2. **A domain boundary acting as a brake.** With intensity honest, the storm arrived hours late (+8.9 h). The cause was geometric: the vortex was feeling the domain-edge zones where the β -plane approximation is tapered off. Enlarging the domain removed the brake—and the storm now arrived hours *early*. The lateness had been masking an overshoot.
3. **A vortex with no outer edge.** The overshoot traced to vortex structure. The analytic wind profile used at initialization decays so slowly with radius that, untruncated, it filled the entire domain; the nominal environmental radius shaped only a dynamically passive pressure integral. The oversized circulation inflated the storm’s β -drift. A cosine taper on the outer wind fixed this—and, in doing so, created the model’s one genuine structural parameter, the taper-onset radius (Section 2.2).
4. **Our best result dissolves.** Hugo’s celebrated 48-km landfall error—the strongest early validation—did not survive the honest vortex: re-run with the structural fixes, Hugo missed by +182 km east under the then-static steering. The 48 km had been a cancellation between an oversized vortex’s inflated westward β -drift and a westward deficit in steering held frozen in time. The repair was time-varying environmental steering, sampled from ERA5 along the model’s own track.
5. **A second flattering number dissolves.** Katrina’s +14-km landfall error repeated the archetype. Scored against verified best-track data, the initialization had placed the storm 98 km west of its true track; an eastward drift of $\sim 1 \text{ m s}^{-1}$ then crossed the observed track at $t \approx 21 \text{ h}$, and the two errors near-cancelled at the landfall hour. Two errors, one flattering number—found only because the reference data themselves were re-verified (chapter 6).
6. **The data layer itself.** Executing the verification that the initialization module’s own documentation demanded revealed that the stored initialization values matched no fix in the HURDAT2 file: positions matching time-shifted interpolants (one a +4.2-h along-track head start), intensities taken from fixes twelve hours later, central pressures matching nothing, one storm filed under the wrong cyclone identifier—and the pipeline’s “observed” reference tracks were synthetic, one of them literally a straight line from the (incorrect) initialization point to landfall. The provenance pattern—plausible, internally consistent, and wrong—is characteristic

of values recalled from memory rather than read from the source. Every downstream score had inherited these references. All initializations and reference tracks were re-derived from the primary file, and the affected results re-scored or retracted (§3.3).

7. **The instruments.** An apparent 13% over-translation—the model seemingly outrunning its own imposed steering—launched the longest hunt of the project, through the advection scheme, the pressure solver, and the damping operators. A Galilean control finally ended it: a balanced vortex in a uniform background flow, which must translate exactly with the flow, instead *appeared* to jitter—because the vortex center-finder reported grid-snapped positions with a ± 1.5 -cell (± 23 km) flicker, a noise floor beneath which the entire family of over-translation measurements had been made. With sub-cell center-finding the control translates faithfully to within 1%, even while self-intensifying, and every over-translation number was retired as instrument artifact. The model’s advection had been faithful all along; the residual that remained, after the instrument was fixed, was the genuine β -drift signal of Section 4.1.

The cascade descended through four strata (Fig. 1): model physics (1), configuration geometry (2, 3), the data layer (5, 6), and finally the measurement instruments themselves (7). Each error was invisible while the ones above it stood. One chapter also ended differently from the others: the outer-wind taper that repaired chapter 3 turned out to *mis-calibrate* the β -drift—too strong and aimed too far poleward—which is not a bug but a physics trade-off, and it was resolved not by tuning to landfall but by calibrating the taper-onset radius against the canonical β -drift band in the testbed (Section 2.2). What remained after that calibration is the bounded, characterized aim residual reported as Result 1 (Section 4.1).

The compensating-errors cascade: each repair exposed the error above it had hidden

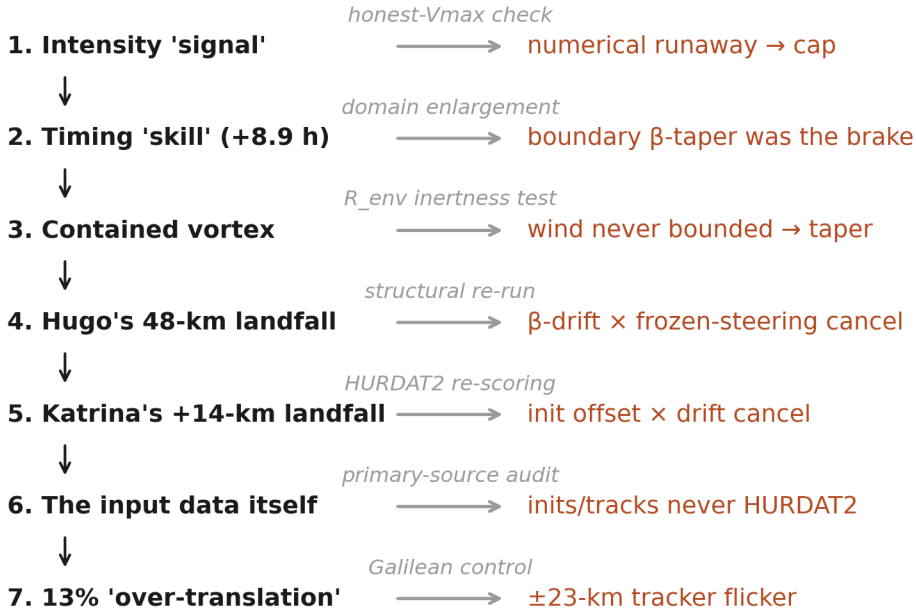


Fig. 1. The compensating-errors cascade of Section 3.2. Each row is a result that looked good (left), the probe that ended it (arrow label), and the error the result had been hiding (right); repairing each exposed the next. The strata descend from model physics through configuration geometry and

the data layer to the measurement instruments themselves.

3.3 The Probes: Retiring Our Own Claims

Clean initialization removes the temptation to tune; it does not by itself protect against over-interpretation. For that we relied on a second habit: subjecting each claim to the cheapest test that could kill it, and registering predictions before runs rather than after. Several claims did not survive, and their retirements shaped the paper more than the results that stood.

Registered predictions. Before re-running the storms with corrected initializations (chapter 6 above), we wrote down predicted landfall bands—Katrina +80 to +130 km east, Hugo +90 to +130 km east—derived from the decomposed error budget, so that the re-runs could confirm or refute the budget rather than merely produce new numbers. Both re-runs landed inside their bands. The residual they exposed was real, coherent—and, as it later proved, still misattributed (below).

Small retractions. An early “13% over-translation” was contaminated by a rotating-frame subtlety in the test harness (an unreferenced background flow undergoes inertial oscillation) before being retired altogether by the Galilean control of chapter 7. A “delayed-onset drift” finding was retracted when re-scored against verified best-track references: the onset had been an artifact of the synthetic reference track, and the drift is in fact steady. An automated sweep-reader once reported a heading change from 350° to 6° as a rotation *toward the northwest*; 6° is east of north. Each of these is small; each would have survived into review unkilld if the claim had not been re-derived from primary data.

The f-plane null. The characterized self-propagation bias (Section 4.1) rests on a control: with the planetary-vorticity gradient set to zero and all else identical, the vortex does not translate at all (residual drift $\lesssim 0.02 \text{ m s}^{-1}$). The drift is genuine β -gyre dynamics, not an advection or tracker bias—the instrument fix of chapter 7 made this null test meaningful.

The two large retirements. The first concerned our own headline. The three discovery storms (Hugo, Katrina, Ivan) showed a tight, same-signed eastward cross-track cluster at landfall, and the manuscript draft claimed it as systematic. Three test storms the configuration had never run—Fran, Michael, and Laura—answered: +8, −99, and −32 km. The cluster was a property of the discovery set, and the claim was retired (Section 4.2). The second concerned the bridge between our two results. If the testbed β -gyre bias were the dominant source of landfall cross-track error, a single bias vector projected through each storm’s landfall geometry should reproduce all six observed errors—a one-afternoon calculation. It fails decisively: it predicts eastward cross-track for every poleward-moving storm, whereas Katrina and Laura approach on the same heading with opposite observed signs (+125 vs −32 km)—no geometry-projected vector can do that. The same test shows where the bias *does* live: it predicts Michael’s along-track error almost exactly (+123 predicted, +124 observed). Steering controls landfall cross-track; the β -gyre bias is real but subdominant, expressed mainly as along-track error on poleward movers (Section 4.2). This cheap test also cancelled an expensive plan: a data-informed vortex-structure initialization scheme, motivated by the bridge claim, was abandoned when the model’s own structure sweeps showed it would push every discovery storm *further* east—the wrong direction.

What this discipline buys is stated most honestly in the negative. The results of Section 4 are not the claims we set out to make; they are the claims that survived. A characterized self-propagation bias whose landfall footprint is bounded and subdominant, and blind track skill under a configuration with no landfall-tuned parameter, are what remained after the cluster claim, the bridge claim, the

over-translation family, and two flattering landfall errors were retired by our own tests. We offer the cascade and the probes as the methodological content of this paper: not that the model is right, but that its errors are where we say they are.

4 Results

Oracle’s track errors are characterized at two levels, and we keep them deliberately separate. The first is a property of the model’s *self-propagation*, isolated in a quiescent-environment testbed in which no environmental steering is imposed. The second is the *landfall track error* of six historical storms run under a single storm-agnostic configuration. The testbed isolates a clean, bounded bias in the model’s β -drift; the landfall error of any individual storm is a convolution of that bias with environmental-steering error and unmodeled track features. As we show below, the characterized β -gyre bias is detectable in the landfall errors—but as a subdominant contribution to along-track timing on poleward-moving storms, *not* as the systematic cross-track displacement that a single-source reading would predict. Landfall cross-track is set by storm-specific steering and is not robust across storm geometry.

4.1 An Over-Rotating β -Gyre: A Self-Propagation Aim Error

Self-propagation by the β -effect is a first-order contributor to tropical-cyclone motion, so we characterize the model’s β -drift directly, in isolation from any imposed steering. We integrate a single balanced vortex on a β -plane with no environmental flow and measure the β -drift vector over the mature window (30–48 h) of the center track. In a representative mature-hurricane configuration (maximum wind 64 m s^{-1} , environmental radius 500 km, wind-profile taper onset at 200 km, intensity cap 70 m s^{-1}), the simulated β -drift is directed approximately $8\text{--}10^\circ$ west of due north—that is, essentially poleward with only a weak westward component—whereas the canonical β -drift of an idealized vortex is oriented toward the northwest quadrant (Holland 1983; Chan and Williams 1987; Fiorino and Elsberry 1989; reviewed by Chan 2005). The discrepancy is thus a *deficit in the westward component* of self-propagation: the model reproduces a β -drift of canonical magnitude ($\approx 2.3\text{--}2.5 \text{ m s}^{-1}$, within the $1\text{--}3 \text{ m s}^{-1}$ range expected for β -drift; Chan 2005) but with a westward component of only $\approx 0.4 \text{ m s}^{-1}$ —under a fifth of the total drift—leaving the self-propagation nearly meridional rather than northwestward (Fig. 2a).

A control integration confirms that this drift is genuinely β -induced rather than a numerical or frame artifact. Repeating the testbed on an f -plane—the meridional gradient of planetary vorticity set to zero, with the same vortex, domain, operators, and time step—produces no systematic translation: the center remains fixed to within the tracker noise floor (drift components $\lesssim 0.02 \text{ m s}^{-1}$) over the full integration, whereas restoring β recovers the $\approx 2.3\text{--}2.5 \text{ m s}^{-1}$ drift. The self-propagation is therefore a true β -drift, and the aim residual is a property of how the model develops the β -gyres, not an advection or center-finding bias.

To determine whether this aim residual is a tunable or numerical artifact, we tested it against the three configuration and discretization levers that could plausibly control it.

Outer-vortex structure. The radius at which the outer wind profile begins to taper sets the horizontal scale of the β -gyres and is the model’s only free outer-structure parameter. In the isolated testbed, reducing the taper-onset radius from 250 to 200 km rotates the β -drift modestly toward the northwest. Applied to a full storm (Ivan), the same change moved the landfall position only $\sim 15 \text{ km}$ —far short of the cross-track residual—with the bulk of the landfall error unchanged. Outer structure

modulates the aim slightly but does not control it; this small landfall footprint already foreshadows the subdominance established in the six-storm audit below.

Subgrid diffusion. The dynamical core employs fourth-order (∇^4) hyperdiffusion with coefficient $\nu_4 = 3.0 \times 10^{11} \text{ m}^4 \text{ s}^{-1}$ for grid-scale noise control. We reduced ν_4 to test the hypothesis that diffusive smearing of the β -gyre asymmetry damps the westward ventilation and so tilts the drift poleward—under which a smaller ν_4 should rotate the aim back toward the northwest. It does not. Reducing ν_4 to $1.0 \times 10^{11} \text{ m}^4 \text{ s}^{-1}$ produced grid-scale energy accumulation (the peak azimuthal wind grew from 42 to 74 m s^{-1} with no corresponding intensification), and on that contaminated solution the mature aim did not rotate toward the northwest—it remained essentially meridional. Further reduction ($\nu_4 \leq 3.0 \times 10^{10} \text{ m}^4 \text{ s}^{-1}$) produced outright numerical divergence. At the operating resolution ($\Delta x = 15.6 \text{ km}$) ν_4 is therefore load-bearing for numerical stability—it is removing genuine grid-scale noise generated by the advection—and is not available as a tuning direction: reducing it neither recovers the westward component nor preserves a clean integration. Diffusion does not control the residual.

Horizontal resolution. Holding ν_4 fixed at its baseline value (which remains below the hyperdiffusion stability limit at all grids tested) and the time step fixed at 30 s (advective Courant number ≤ 0.27 at the finest grid, bounded by the 70 m s^{-1} intensity cap), we refined the horizontal grid by a factor of two, from $\Delta x = 15.6$ to 7.8 km. The mature β -drift heading is invariant to within 2° across the refinement, while the magnitude converges monotonically (Table 1). The aim residual is therefore not a discretization artifact: the β -gyre is adequately resolved at the operating resolution, the drift *magnitude* is grid-converged, and additional refinement does not rotate the aim toward the expected northwest orientation.

Table 1. Mature (30–48 h) β -drift in the quiescent-environment testbed as a function of horizontal resolution, with ν_4 and Δt held fixed. Heading is measured clockwise from due north (i.e., $350^\circ \approx 10^\circ$ west of north); a westward rotation toward the expected northwest β -drift would appear as a *decrease* in heading.

Grid (n)	Δx (km)	β -drift heading	Δ vs. coarsest	β -drift speed (m s^{-1})
320	15.6	350°	—	2.49
480	10.4	351°	$+1^\circ$	2.39
640	7.8	352°	$+2^\circ$	2.29

Two further diagnostics identify the mechanism as a β -gyre that over-rotates past its equilibrium orientation rather than settling onto it. First, the anomaly is *intensity-independent* in its cross-track-relevant component. Varying the vortex strength over a threefold range, the westward component of the β -drift remains nearly constant at $\approx 0.4 \text{ m s}^{-1}$ —well short of the northwestward propagation expected of canonical β -drift—while the northward component, and with it the total drift speed, scales with maximum wind as expected. The drift vector therefore rotates poleward as intensity increases (its mature heading shifts from $\approx 338^\circ$ at the weakest vortex to $\approx 351^\circ$ at the strongest), but this reflects a growing poleward component acting over a near-fixed westward one, not a change in the underlying anomaly: the westward component does not recover with intensity. An aim error set by the storm’s own swirl or vertical shear would scale with intensity; a westward component that stays small regardless points instead to the β -Rossby gyre dynamics themselves. Second, resolving the drift heading in time shows that it does not lock onto the canonical orientation and remain there. The heading begins near northwest early in the integration and climbs steadily poleward

thereafter (Fig. 2b)—by $\approx 14^\circ$ over 48 h at the strongest vortex and $\approx 16^\circ$ at the weakest—so that the *rate* of this poleward precession, like the smallness of the westward component, is insensitive to intensity. The β -gyre asymmetry continues to rotate cyclonically past the orientation at which a correctly equilibrated gyre would balance, so that the time-mean self-propagation is aimed too far poleward; both intensity-invariant signatures—the persistently small westward component and the intensity-independent precession rate—point to β -Rossby gyre dynamics rather than a swirl-driven mechanism. Figure 3 shows the corresponding $m = 1$ vorticity asymmetry: its amplitude saturates while its orientation holds poleward of northwest—the structural signature of this equilibration failure.

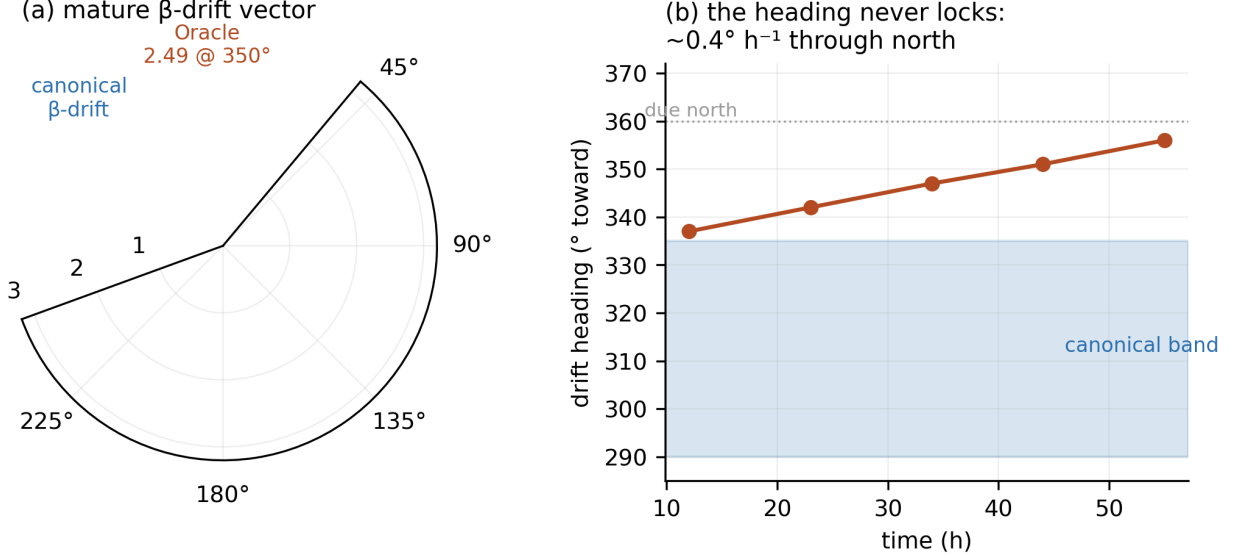


Fig. 2. The testbed β -drift diagnosis. (a) The mature (30–48 h) drift vector at the production configuration: canonical magnitude (2.49 m s^{-1}) aimed at 350° , nearly due poleward, against the canonical northwest β -drift band (shaded). (b) Drift heading versus time: the heading never locks, precessing poleward at $\approx 0.4^\circ \text{ h}^{-1}$ through—and past—due north; the shaded band is the canonical range.

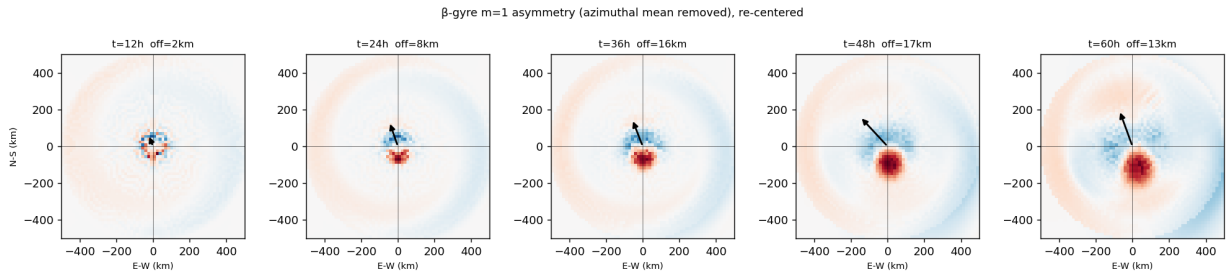


Fig. 3. β -gyre $m = 1$ vorticity asymmetry (azimuthal mean removed, vortex re-centered) at $t = 12\text{--}60$ h in the quiescent-environment testbed. The asymmetry intensifies as the storm drifts; the black arrow is the swirl-removed steering flow, which matches the simulated β -drift. The gyre amplitude saturates while its orientation holds poleward of the canonical northwest—the structural signature of the equilibration failure.

The aim residual survives all three levers: it is insensitive to outer-vortex structure, it cannot be diffused away without loss of numerical stability, and it is invariant under a doubling of horizontal

resolution. We therefore characterize it not as a tunable bias or a discretization error but as an intrinsic property of the model’s β -gyre dynamics—a bounded, systematic poleward bias in self-propagation aim, expressed as a deficient westward component of the β -drift, arising because the simulated β -gyres over-rotate past their equilibrium orientation. This is a characterized *model property*, isolated in the testbed and independent of any storm; what it does to a real landfall is a separate, falsifiable question, addressed next.

4.2 Landfall Track Errors Across Six Storms

We evaluate the configuration against six historical landfalls—Hugo (1989), Katrina (2005), Ivan (2004), Fran (1996), Michael (2018), and Laura (2020)—under one storm-agnostic configuration. The configuration carries *no landfall-tuned parameter*: its single outer-structure parameter, the taper-onset radius, is calibrated in the quiescent testbed to the canonical β -drift band (Section 2.2), independent of every storm’s track and landfall. All six storms are therefore track-*validation* cases, not calibration targets. Hugo, Katrina, and Ivan are the three discovery storms behind the eastward cross-track cluster whose retirement Section 3.3 recounts; Fran, Michael, and Laura were added to test whether that behavior generalizes to storms the model had never been run on.

We report both verification layers of Section 2.4 for every storm (Table 2): the along-track/cross-track decomposition relative to the observed best track, evaluated at the observed landfall fix, and the same-latitude threshold comparison. The decomposition is the authoritative cross-track reading. Michael illustrates why: its true error is a +124-km along-track overshoot (the model runs poleward while the observed storm recurves northeast), and because the observed motion is turning, that overshoot reprojects onto the same-latitude axis as a spurious ~ 75 km of westward cross-track. The decomposition renders the error legible as timing and removes the artifact.

Table 2. Landfall track error for six storms under the single storm-agnostic configuration. Along-track/cross-track components are evaluated at the observed landfall fix (positive along-track = model ahead of observed; positive cross-track = model to the right of, i.e. approximately east of, the observed track). The same-latitude layer reports the longitude offset and crossing-time difference at the observed landfall latitude. Landfall heading is the observed storm motion, clockwise from north.

Storm	Role	Along (km)	Cross (km)	Same-lat offset (km)	Crossing Δt (h)	Heading	Translation
Hugo	discovery	+23	+110	+103	−2.3	$\sim$ 325°	moderate
Katrina	discovery	+77	+125	+114	−2.6	$\sim$ 350°	moderate
Ivan	discovery	+249	+126	+68	−8.1	$\sim$ 340°	recurver
Fran	test	−45	+8	+23	+1.2	$\sim$ 335°	fast (~ 15 kt)
Michael	test	+124	−99	−75	−5.2	$\sim$ 010°	slow (~ 11 kt)
Laura	test	+37	−32	−39	−1.3	$\sim$ 350°	moderate (~ 13 kt)

The three discovery storms all show a large eastward (right-of-track) cross-track error, +110 to

+126 km—the tight, same-signed cluster of Section 3.3. *It does not generalize* (Fig. 4a). The three test storms show cross-track errors of +8, −99, and −32 km—two of them reversed in sign and none of comparable magnitude. The eastward cross-track is therefore not a robust property of the model: across storm geometries it is neither same-signed nor clustered. (Because the configuration has no landfall-tuned parameter, this is not an in-sample/out-of-sample overfitting story; it is a statement about the *geometry-robustness* of the error.)

The decisive evidence that landfall cross-track is steering-controlled rather than bias-controlled is a within-pair comparison. Katrina and Laura make landfall in nearly the same location (the central Gulf coast) on nearly the same heading ($\approx$ north), yet their cross-track errors are opposite in sign, +125 and −32 km. A track error produced by a single self-propagation bias, projected through a common heading, *cannot* change sign between two such storms; the difference must arise from their differing environmental steering. Storm-specific steering, not the characterized β -gyre bias, sets the sign of the landfall cross-track.

To test directly whether the characterized β -gyre bias is the dominant landfall-error source, we projected the testbed aim-error vector through each storm’s observed landfall motion and transit time, predicting the along/cross split each storm would show if that bias dominated. Because all six storms make landfall on roughly poleward headings, and the model’s self-propagation is westward-deficient (i.e., biased eastward relative to canonical), the projection predicts an eastward cross-track for *all six* storms (Fig. 4b). It reproduces the discovery three within $\sim 25\%$ and recovers the along-track error of the poleward-moving storms—notably Michael’s early-arrival overshoot, +123 km predicted against +124 km observed—but it predicts the wrong sign for every test-storm cross-track. The characterized bias is therefore detectable in the landfall errors, but it manifests as a subdominant contribution to along-track timing on poleward movers, not as a controlling cross-track displacement.

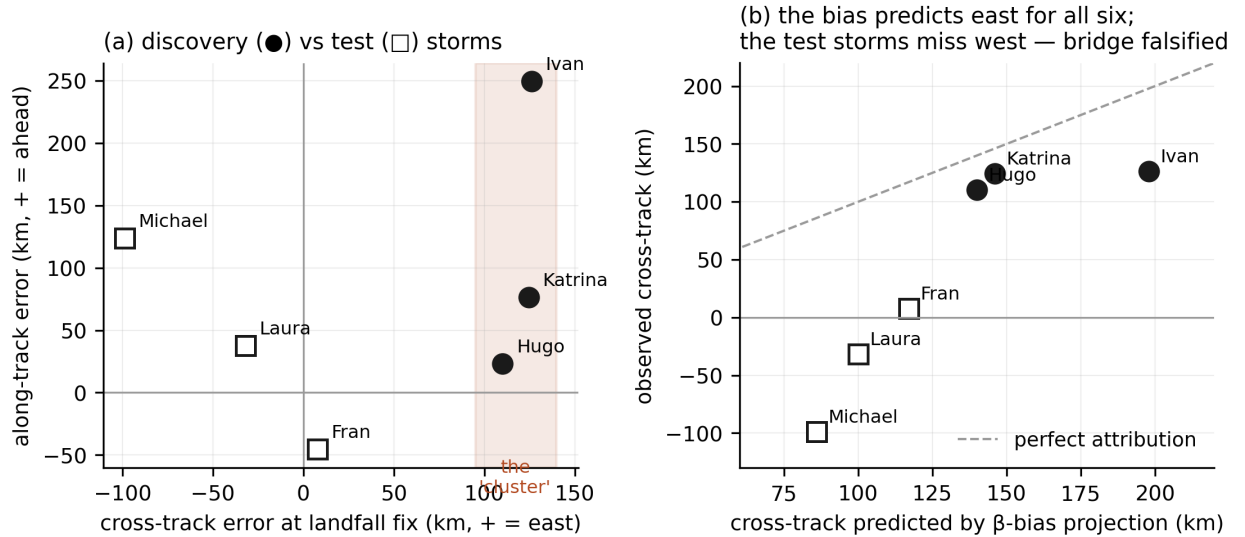


Fig. 4. The six-storm landfall record and the falsified bridge. (a) Along-track versus cross-track error at the observed landfall fix for the discovery (filled) and test (open) storms; the shaded band marks the discovery storms’ eastward cluster, which the test storms break. (b) Observed cross-track error versus the cross-track predicted by projecting the testbed β -drift bias through each storm’s landfall geometry: the projection predicts east for all six storms, the test storms miss west or near-zero, and no single bias vector can reproduce the record (dashed line: perfect attribution).

We accordingly report two separate facts and no bridge between them. First, the model’s β -gyre over-rotates poleward, producing a bounded, characterized self-propagation aim bias—a model property, isolated in the testbed and independent of any storm. Second, the configuration reproduces six historical landfalls—none of them a calibration target, since it carries no landfall-tuned parameter—with along-track and cross-track errors of tens of kilometers and characterized timing errors. These landfall errors are dominated by environmental-steering error and unmodeled recurvature; the characterized β -gyre bias is a detectable but subdominant along-track term, consistent with but not the cause of the landfall cross-track. We make no claim that the testbed bias produces a systematic landfall cross-track displacement—the six-storm audit shows that it does not. The empirical strength of the configuration is its *blind track skill*: with its single outer-structure parameter calibrated to canonical β -drift rather than to any landfall, it places six storms—three of them never previously run—within tens of kilometers to ~ 125 km cross-track, with no landfall-fitted parameter to absorb the error. This is consistent with the study’s broader methodology, in which clean initialization is used to expose true model residuals rather than to conceal them behind compensating errors (Section 3).

5 Discussion

5.1 What the Over-Rotation Is—And Is Not

The testbed diagnosis of Section 4.1 can be stated in one sentence: the model’s β -gyres reach the right amplitude and the wrong orientation, and the orientation never locks. Drift speed saturates near the canonical magnitude while the drift heading precesses steadily poleward through—and past—due north, at a rate insensitive to vortex intensity. In the canonical picture, the gyre pair equilibrates where the vortex circulation’s advection of the gyres balances their generation by the β -effect, and the storm settles onto a steady northwestward bearing (Fiorino and Elsberry 1989; Chan and Williams 1987). In our model the amplitude side of that balance is achieved and the phase side is not: whatever arrests the cyclonic winding of the gyre pair in nature—and in the models that reproduce it—is too weak here.

We know three things it is *not*. It is not an advection or measurement artifact (the f-plane null and the validated instrument); it is not diffusive smearing (reducing ν_4 degrades the solution without rotating the aim); and it is not under-resolution in any simple sense (the heading is flat across a doubling of resolution). Three candidates remained, untested at the time of this analysis, all cheap to test: the imposed outer-wind cutoff, which sits inside the radial annulus where the gyres live, and whose sweeps move the aim only to a floor; the reduced barotropic configuration, since a vortex with vertical structure disperses β -Rossby energy differently than a barotropic one; and the specific shape—rather than the radius—of the taper. A subsequent profile-family test has since implicated the first: replacing the compact-support taper with a smooth outer envelope recovers canonical, phase-locked β -drift in the testbed, and the mechanism and its six-storm validation are reported in a companion paper (in preparation).

5.2 What a Landfall Can Attribute—And a Test Worth Exporting

The six-storm record enforces a discipline on attribution that we suspect generalizes. A landfall error convolves self-propagation error, steering error, and unrepresented track features; landfall data alone cannot deconvolve them, and Section 4.2 shows how persuasive the resulting illusions can be—three discovery storms produced a 16-km-tight eastward cluster that was, nonetheless, not a property of the model. The instrument that caught this cost one afternoon: take the candidate bias

vector, project it through each storm’s landfall geometry, and demand it reproduce the held-out storms’ along/cross splits. We commend the projection test as a general gate. Any proposed model change that is motivated by a landfall residual—a structural initialization scheme, in our case—can be evaluated on paper against held-out geometry before a line of it is built. Ours failed the gate; the sweeps then showed the fix would have moved every discovery storm the wrong way. The expensive experiment was cancelled by the cheap one, which is the correct order.

5.3 How Much of the Skill Belongs to the Steering?

An honest accounting of the blind track skill must start with what the configuration is given: ERA5 is a reanalysis, so the environmental flow the model relaxes toward is close to the truth. The skill demonstrated here is therefore hindcast skill under near-true steering—evidence about the *model’s dynamics and the method*, not a claim of operational forecast value, where steering must itself be forecast. Within that frame, the attribution of Section 4.2 says the steering carries the cross-track outcome and the model’s contribution is to integrate it faithfully while adding a bounded, characterized self-propagation.

Two cheap measurements, made after external readers pressed the point, calibrate that statement. A persistence baseline—each storm’s observed motion at initialization, extrapolated over its transit—misses the landfall position by more than the model on all six storms (mean total error 225 km against the model’s 132), with the deficit concentrated in timing: every storm in this sample accelerated poleward, and persistence cannot know that. Its cross-track error alone, however, is comparable (117 km RMS against the model’s 95), so the model’s edge over a no-skill extrapolation lives in landfall position and timing rather than in cross-track placement—the same division of labor the attribution asserts. And a steering-only tracer—a point advected by the sampled annulus DLM, with no vortex at all—reproduces the cross-track record nearly as well as the full model (105 km RMS) while arriving late on every storm; the model runs ahead of the tracer along-track on all six, by 104 to 261 km, which is the β -drift’s poleward push measured directly. The steering carries the cross-track; the vortex’s self-propagation carries the timing.

Two caveats on the steering pathway remain, deferred here from Section 2.3. First, the 3–7° annulus average is not a storm-removed environmental field; at those radii it retains some of the storm’s own outer circulation, and the sensitivity of track to annulus width—or to storm-removed or time-lagged steering—had not been tested at the time of this analysis. Second, and more subtly, a deep-layer mean drawn from reanalysis in a storm-centered annulus may already contain part of the storm’s *propagation*, not just its environment; if so, the model’s own β -drift partially double-counts it, and the symptom would be systematic poleward over-translation on storms whose DLM tracks their motion—a signature at least consistent with our largest timing miss (Ivan, –8 h). Both are bracketing experiments, not rebuilds; both have since been run, and are reported in the companion paper.

5.4 One Model, or a Class?

We do not know whether the over-rotating β -gyre is a property of this model or of a family of models that share its ingredients—soundproof cores, bounded vortices, hyperdiffusive closures, or simply β -gyres asked to equilibrate at 15-km resolution. The diagnostic, however, is portable and cheap: a quiescent β -plane, a balanced vortex, a mature-window drift vector, and an f-plane null. Any track model can be asked the same question in an afternoon of compute, and the answer is a vector, not an interpretation. We would be glad to learn whether other cores over-rotate; either

answer is informative, and the comparison would locate the mechanism faster than we can from inside one model.

5.5 What Blind Skill Demonstrates, and What Would Strengthen It

Six storms—three of them never run during development—are enough to *retire* a false claim, which is the use we made of them; they are thin support for a strong positive claim, and we state the skill accordingly. Strengthening it is now mechanical rather than conceptual: the unified pipeline reduces a new storm to a registry entry and a reanalysis download, and the storms that would stress the method most are known from the present six—more fast straight-movers and slow recurvers, and a second basin with different climatological steering. Beyond track, the core’s thermodynamic pathway—validated but switched off throughout this paper—is the natural frontier: with buoyancy active, intensity becomes a prognostic quantity rather than a capped one, and the clean-initialization discipline would face a harder test, since intensity error and track error feed each other through exactly the kind of compensation this paper is about. We intend to arrive at that test with the same rule we arrived here with: the model is never granted the benefit of the doubt.

6 Conclusions

We set out to characterize a tropical-cyclone model’s track error without giving the model any opportunity to flatter us, and we report what survived. Two facts stand. First, the model’s β -gyres over-rotate: in isolation its storms propel themselves with canonical β -drift magnitude but nearly poleward aim, a westward deficit that survives structure, diffusion, and resolution levers, an f-plane null, and a validated measurement instrument. It is a bounded, reproducible, mechanistically located model property, and we have left it in the model, characterized rather than concealed. Second, the same configuration—one set of coefficients, one structural parameter calibrated to idealized-vortex physics, nothing anywhere adjusted against a landfall—reproduces six historical landfalls, including three storms it had never run, with cross-track errors from a few tens of kilometers to roughly 125 km.

The relation between these facts is the caution at the center of this paper. The obvious manuscript connects them—a poleward-biased self-propagation pushes poleward-moving storms east, and our first three storms all missed east by a strikingly consistent margin. That manuscript was drafted, and it was wrong: the three test storms broke the cluster, and a single projection calculation showed that no fixed self-propagation bias can reproduce six landfall geometries in which two storms on the same heading miss on opposite sides. Steering controls landfall cross-track in these cases; the characterized bias is detectable—it predicts the along-track error of the slowest poleward mover almost exactly—but subdominant. We publish the two facts separately because that is what the evidence supports, and we publish the falsification because it is the most instructive result we have.

That instruction is the third contribution. Under clean initialization, every flattering number we ever produced eventually dissolved into a pair of canceling errors—seven times, in strata descending from model physics through configuration and input data to the measurement instruments themselves—and every dissolution was forced by a cheap, prespecified test: registered prediction bands, primary-source verification, null controls, a Galilean instrument check. None of this required new theory or new computing; it required only that the model never be granted the benefit of the doubt. A model whose errors are characterized, bounded, and honestly placed is more useful—to forecasters weighing its guidance, to developers hunting its next error, and to anyone deciding how far to trust it—than a model whose errors have been arranged to cancel. Six hurricanes suggest

that standard is attainable without sacrificing skill. We offer the discipline, as much as the model, as this paper’s contribution.

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- [Companion paper, in preparation: the compact-support mechanism, the envelope fix, and its six-storm validation.]